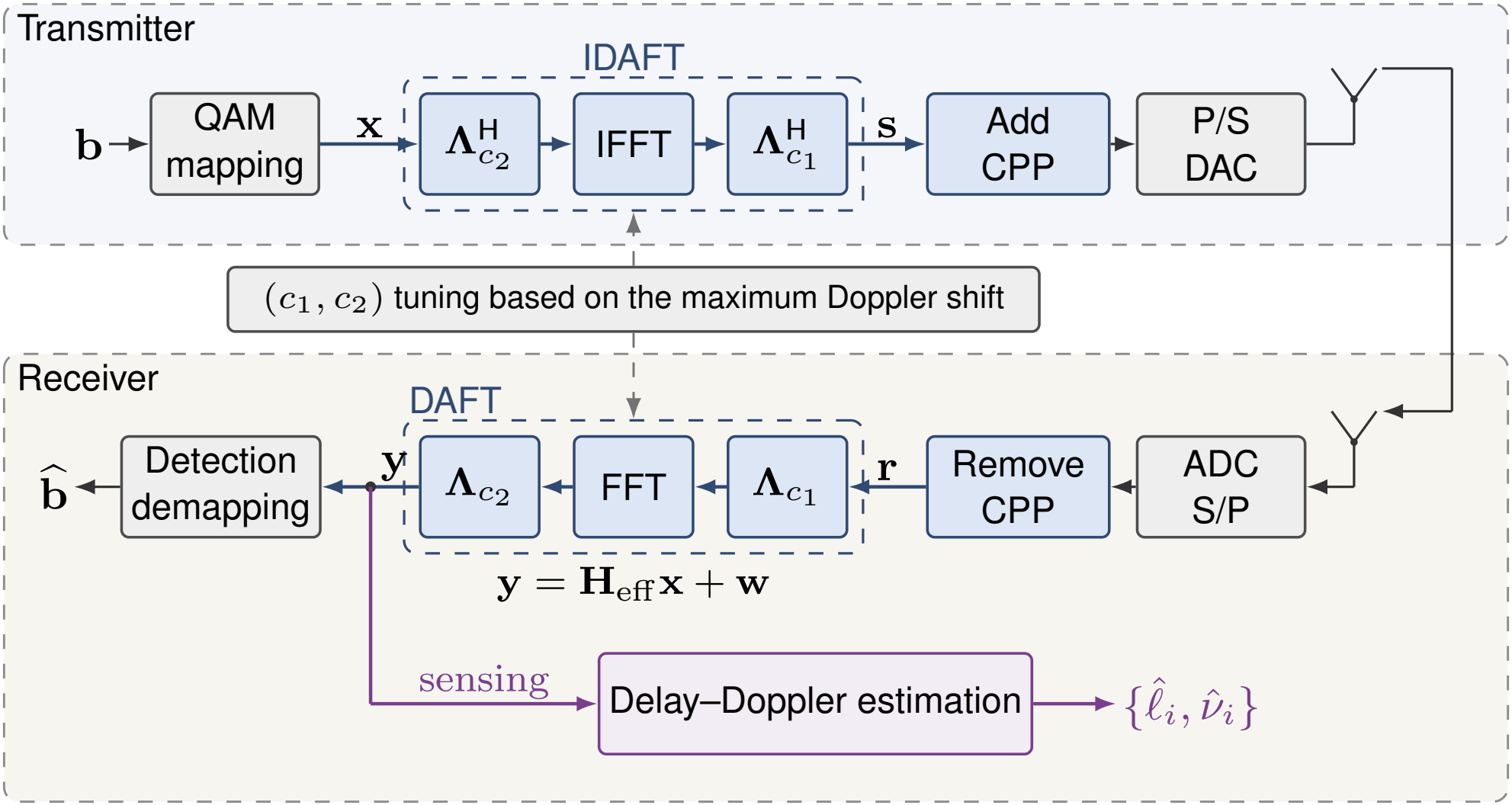

Transmitter
IDAFT
b
QAM mapping
x
$\Lambda_{c_2}^{H}$
IFFT
$\Lambda_{c_1}^{H}$
s
Add CPP
P/S DAC
(c_1, c_2) tuning based on the maximum Doppler shift
Receiver
DAFT
$\widehat{\mathbf{b}}$
Detection demapping
y
Λ_{c_2}
FFT
Λ_{c_1}
r
Remove CPP
ADC S/P
$\mathbf{y} = \mathbf{H}_{\text{eff}}\mathbf{x} + \mathbf{w}$
sensing
Delay–Doppler estimation
$\{\hat{\ell}_i, \hat{\nu}_i\}$